Software Requirements Specification (SRS)

VeriSync: Secure Placement Assessment Platform

Version 1.1 (Revised Baseline) | **Date:** 24 October 2026 | **Status:** Proposed Baseline for Review

Revision 1.1 retains all requirements of v1.0 and adds system diagrams, actor and use-case specifications, sequence and state models, API payload examples, sample test cases, and a clearer requirements structure.

Document Control

Field	Details
Project Name	VeriSync: Secure Placement Assessment Platform
Document	Software Requirements Specification
Version	1.1 (Revised)
Prepared For	Internship Project
Intended Users	Super Admins, Placement Administrators, Candidates, Technical Evaluators
Candidate Environment	Installed Windows desktop application under managed kiosk restrictions
Administrator Environment	Web-based dashboard
Backend	Cloud-hosted application services and relational database
Primary Operating Environment	Windows lab computers

Revision History

Version	Date	Description
1.0	24 October 2026	Initial comprehensive SRS baseline
1.1	24 October 2026	Added diagrams (context, architecture, ER, state, sequence), use-case specifications, API examples, sample test cases, requirement conventions, and an expanded glossary

Team

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Requirement Conventions

- **Must:** mandatory for the initial production release.
- **Should:** strongly recommended; expected unless formally deferred.
- **Could:** optional enhancement for a later release.
- Requirement IDs follow the pattern `FR-<AREA>-<NNN>` (functional) and `NFR-<AREA>-<NNN>` (non-functional). Business rules use `BR-<NNN>`. Use cases use `UC-<NN>`.
- Where a requirement has no explicit priority, it is **Must** unless it uses the word "should" or "could".

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1. Introduction

1.1 Purpose

This document defines the functional, non-functional, technical, operational, and security requirements for **VeriSync**, a secure assessment platform for technical placement recruitment and candidate evaluation.

It is the common reference for developers, testers, supervisors, system administrators, placement coordinators, and other stakeholders. It states what the system must do, how its major components interact, how security is enforced, and how success is verified.

The SRS shall be used to:

- Guide architecture and implementation.
- Fix the scope of the initial release.
- Define responsibilities and permissions for each user type.
- Specify assessment creation, question management, timing, submission, and scoring.
- Specify coding and SQL assessment and automated evaluation.
- Define candidate-application lockdown requirements.
- Define data protection, audit logging, and cloud security requirements.
- Guide integration, security, and acceptance testing.
- Reduce ambiguity during development and maintenance.

1.2 Project Overview

VeriSync is a centralized platform for campus placement drives, technical recruitment examinations, and skill-based candidate evaluations. It has three primary components:

1. **Super Admin Web Dashboard:** manages candidates, questions, assessments, assignments, policies, results, and configuration.
2. **Candidate Desktop Application (Lockdown Client):** installed on lab computers; used by candidates to complete assigned assessments.
3. **Cloud Backend and Database:** authentication, authorization, question delivery, timing, answer processing, evaluation, reporting, and audit logging.

Candidates receive credentials from an authorized administrator. They cannot create accounts, change assignments, view answer keys, or modify examination settings.

Supported formats: aptitude questions, technical MCQs, SQL problems, debugging exercises, and programming challenges.

1.3 Objectives

ID	Objective
OBJ-01	Provide a secure, centralized platform for technical placement assessments.
OBJ-02	Ensure only authorized candidates can access their assigned assessments.
OBJ-03	Let the Super Admin manage candidates and assessment configuration.
OBJ-04	Enforce per-question timing and optional overall deadlines on the server.
OBJ-05	Prevent advancing without submitting the current question, except under timeout or authorized recovery policy.
OBJ-06	Auto-submit expired questions and completed assessments by defined rules.
OBJ-07	Reduce cheating opportunities through kiosk restrictions, application controls, and audit logging.
OBJ-08	Support programming assessments in multiple languages.
OBJ-09	Automatically evaluate objective questions and eligible coding submissions.
OBJ-10	Generate reliable scores, rankings, reports, and records.
OBJ-11	Protect candidate information and examination content from unauthorized access.
OBJ-12	Support secure centralized administration across many lab computers.

1.4 Scope

1.4.1 In Scope

- Super Admin authentication and authorization.
- Candidate account creation, modification, deactivation, and password management.
- Candidate profile management.
- Assessment creation, scheduling, assignment, and configuration.
- Question bank management and multiple question formats.
- Per-question timers and optional overall timers.
- Sequential candidate navigation, automatic answer submission, advancement, and final submission.
- Server-side scoring.
- Coding questions and isolated code execution; SQL questions and isolated SQL execution.
- Role-based placement assessment templates.
- Candidate session management.
- Lab computer registration and client-version monitoring.
- Kiosk and lockdown integration.
- Audit logging and security-event recording.
- Result management, candidate and assessment reports.
- Secure cloud deployment, backup, recovery, monitoring, and maintenance.

1.4.2 Out of Scope (Initial Release)

- Facial recognition and biometric identification.
- AI-based proctoring and automated behavioural judgments.
- Live video surveillance.
- Automated recruitment decisions based solely on AI.
- Integration with external applicant tracking systems.
- Mobile candidate examinations.
- Fully offline assessments.
- Integration with university student information systems.
- Automated email or SMS credential delivery without an approved workflow.

- External identity-provider integration.
- Payment processing.

The architecture shall allow future enhancements without compromising the security of the initial release.

1.5 Intended Audience

Project supervisors and mentors, software developers (backend, frontend, desktop), database administrators, cloud engineers, security and QA testers, placement and recruitment administrators, and authorized evaluators.

1.6 Definitions and Abbreviations

Term	Definition
SRS	Software Requirements Specification
RBAC	Role-Based Access Control
API	Application Programming Interface
MFA	Multi-Factor Authentication
MCQ	Multiple-Choice Question
SQL	Structured Query Language
TLS	Transport Layer Security
RPO / RTO	Recovery Point Objective / Recovery Time Objective
Kiosk Mode	A managed OS environment restricting a device to approved applications or workflows
Candidate	A person assigned an assessment
Super Admin	The highest-privilege platform administrator
Assessment	A configured examination containing questions and evaluation rules
Question Bank	The centrally managed repository of questions

Term	Definition
Question Attempt	A candidate's interaction with one question
Auto-Submission	Server-controlled submission when a deadline expires
Audit Log	A timestamped record of a relevant system or administrative event
Hidden Test Case	A coding evaluation input not disclosed to candidates
Coding Sandbox	An isolated environment for compiling and executing candidate code
Assessment Session	The server-managed record of one candidate attempt
Lockdown Client	The installed application used to take assessments under managed restrictions
Idempotency Key	A client-generated token that makes repeated identical requests safe
Correlation ID	An identifier linking all log entries produced by one request or operation
Provisional Result	A result issued before all manual evaluation is complete

2. Overall Description

2.1 Product Perspective

VeriSync is a cloud-hosted client-server system. The Super Admin uses a web browser; candidates use an installed desktop application on designated lab computers. Both communicate with the backend through authenticated, authorized APIs over HTTPS.

The backend holds all authoritative assessment data. The candidate application displays questions, collects responses, and relays candidate actions, but it is **never trusted** to determine deadlines, correct answers, scores, or authorization. The coding execution service is isolated from the main backend to contain the risk of running untrusted code.

2.2 Logical Components

1. Admin Web Application
2. Candidate Desktop Application
3. Authentication and Session Service
4. Candidate Management Service
5. Assessment Management Service
6. Question Delivery Service
7. Timing and Submission Service
8. Answer Processing Service
9. Evaluation and Scoring Service
10. Coding Execution Service
11. Results and Reporting Service
12. Audit and Monitoring Service
13. Relational Database
14. Private Object Storage (where required)
15. Cloud Infrastructure and Deployment Services

The database shall not be directly reachable from candidate computers. The coding execution service shall not have unrestricted access to the application database, cloud credentials, internal services, or external networks. Section 3 gives the architecture diagrams.

2.3 User Classes

Class	Responsibilities	Restrictions
Super Admin	Create and manage candidates, questions, and assessments; configure timers and marking; assign assessments; manage schedules; review results and audit logs; manage lab devices and system configuration; control account status	Requires strong authentication (MFA for production); access limited to authorized personnel
Candidate	Sign in with issued credentials; view assigned assessments and instructions; start when eligible; answer and submit; view remaining time; receive timeout and submission notices; view own results only if policy permits	Cannot self-register, access others' records, manage questions, view answer keys, change timing, or alter assignments
Technical Evaluator (optional, future or separately enabled)	Review subjective answers; evaluate coding submissions manually; assign marks by rubric; add comments	No access to candidate account management or global configuration unless granted
Lab Staff (operational)	Run the examination session in the lab; perform documented recovery for technical failures	No access to answers, keys, or scores; actions are logged

In the initial release the Super Admin may perform all evaluation-related operations.

2.4 Operating Environment

Candidate environment: designated lab computers; installed Windows application; managed kiosk configuration; network connectivity to the cloud backend; supported security updates; approved application version; restricted candidate OS account.

> **Windows 10 deployment constraint:** standard Windows 10 support ended on 14 October 2025. Before deployment the organization must verify each lab computer's edition, Extended Security Updates coverage, and security status. Migration to a supported Windows version is recommended.

Administrator environment: supported desktop browser; internet connectivity; authorized administrator account; HTTPS access to the dashboard; MFA for privileged accounts.

Cloud environment: managed application hosting; managed relational database; HTTPS certificates; private storage; centralized logging and monitoring; automated backups; network restrictions and access controls.

2.5 Assumptions

1. The organization will provide suitable cloud infrastructure.
2. Candidates will use institution-managed lab computers.
3. Each candidate receives unique credentials.
4. Internet access is available during assessments.
5. Questions and marking rules are configured by authorized personnel.
6. The organization will define policies for interruptions, candidate privacy, and result retention.
7. The coding engine will be separately configured and secured.
8. Peak simultaneous candidates will be determined before load testing.
9. A supported backend runtime and database version will be used.
10. Device lockdown is enforced through operating-system management in addition to application-level restrictions.

2.6 Constraints

- The candidate application must support the selected Windows desktop environment.

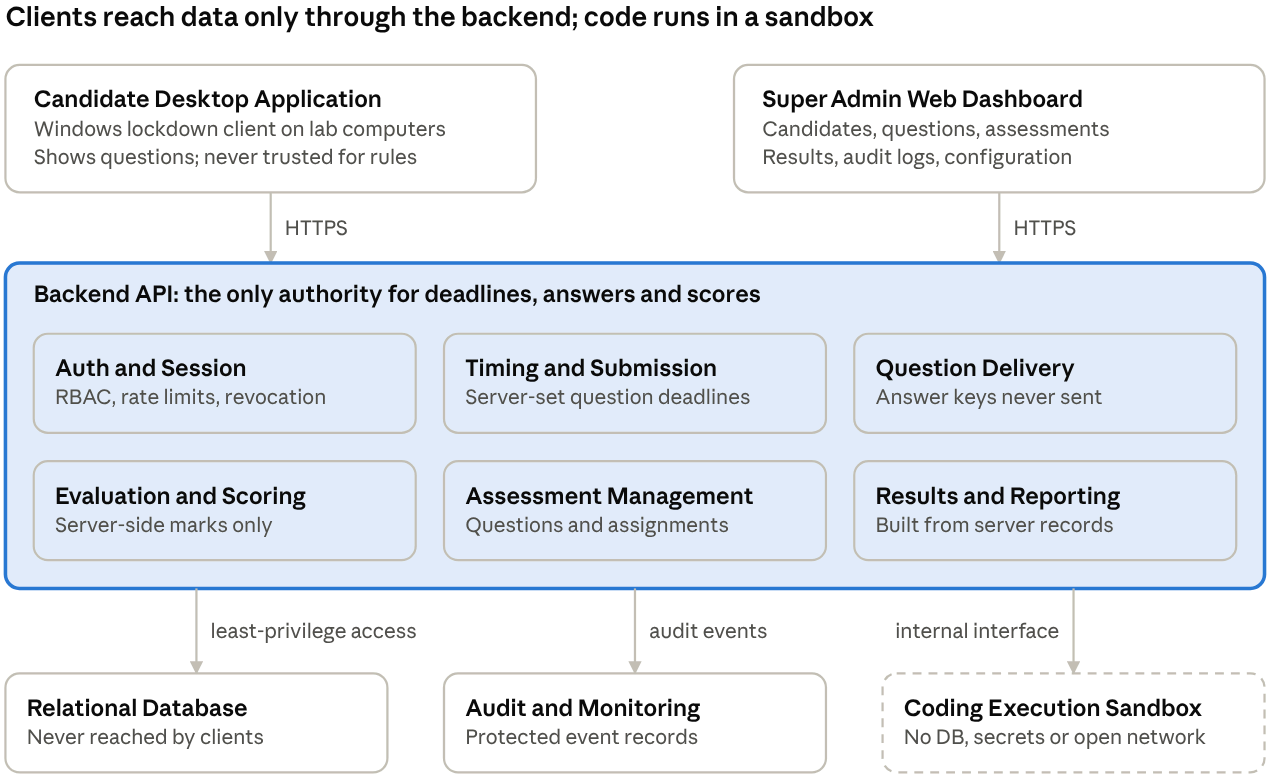
- The backend must be reachable securely over the internet.
- Assessment timing must be controlled by the server and must not depend on the candidate's local clock.
- Candidate permissions must be enforced independently of the user interface.
- Untrusted candidate code must run in an isolated environment, with no unrestricted internet access from the sandbox.
- Full-screen mode alone must not be treated as a security guarantee.
- Cloud hosting and code-execution costs must be considered in deployment planning.
- Final performance targets must be validated against the expected examination load.

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3. System Models and Diagrams

This section is new in v1.1. It gives the architecture, timing flow, state models, and data relationships that the functional requirements in Section 5 build on.

3.1 System Architecture



system architecture · 2 clients, 6 services, 3 back-end stores

Both clients talk only to the backend API over HTTPS. The database, audit service, and coding sandbox sit behind the API; the sandbox has no route to the database, cloud credentials, or application secrets.

Component	Responsibility	Trust Boundary
Candidate Desktop Application	Login, question display, answer entry, timer display, code editor, reconnection handling	Untrusted: any rule it enforces is re-checked by the server
Admin Web Dashboard	Candidate, question, assessment, schedule, result, and audit management	Authenticated and authorized per request; MFA for privileged users
Backend API	Authentication, authorization, timing, answer acceptance, scoring, reporting	Trusted authority
Relational Database	Durable system of record	Reachable only from backend services with least-privilege credentials
Audit and Monitoring	Event history and operational health	Write-protected from application users
Coding Execution Sandbox	Compile and run candidate code and SQL under limits	Fully isolated; no production credentials

3.2 Per-Question Timing Flow

The server sets every deadline and rejects late answers



Steps repeat for every question. The client timer is for display only.

per-question timing flow · 6 steps, 1 decision

A question's deadline is the server-recorded start time plus its configured limit, never later than the overall assessment deadline. A submission that arrives after the deadline is rejected and the question is finalized as timed out under the configured auto-advance policy.

3.3 Assessment Session States

State	Meaning	Allowed Next States
Not Started	Assignment exists; candidate has not begun	Ready, Cancelled
Ready	Eligibility and instructions acknowledged	Active, Cancelled
Active	Server clock is running; one question is active	Interrupted, Submitted, Expired
Interrupted	Connection or client lost; deadlines keep running	Active, Expired, Cancelled

State	Meaning	Allowed Next States
Submitted	Final submission accepted; no further edits	none (terminal)
Expired	Overall deadline passed; the server finalized the attempt	none (terminal)
Cancelled	Withdrawn by an authorized administrator, logged with reason	none (terminal)

The backend rejects any transition not listed. Reopening or extending an attempt is an audited administrative override (BR-012).

3.4 Question Attempt States

State	Meaning	Allowed Next States
Pending	Not yet reached in sequence	Active
Active	Accepting answers until its deadline	Submitted, Timed Out
Submitted	Answer accepted and acknowledged	Evaluation Pending, Evaluated
Timed Out	Deadline passed; configured timeout rule applied	Evaluation Pending, Evaluated
Skipped	Only where the assessment allows skipping	Evaluation Pending, Evaluated
Evaluation Pending	Awaiting automatic or manual evaluation	Evaluated
Evaluated	Marks awarded	none (terminal)

In sequential mode only the Active attempt accepts a new answer.

3.5 Data Relationships

Parent	Relationship	Child
users	1 to 0..1	candidate_profiles
candidate_profiles	1 to many	assessment_assignments
assessments	1 to many	assessment_assignments
assessments	1 to many	assessment_questions
questions	1 to many	assessment_questions
questions	1 to many	question_options
questions	1 to many	test_cases
assessment_assignments	1 to many	assessment_sessions
assessment_sessions	1 to many	question_attempts
assessment_questions	1 to many	question_attempts
question_attempts	1 to many	answers
question_attempts	1 to many	coding_submissions
assessment_sessions	1 to 0..1	results
users	1 to many	audit_logs (as actor)

`lab_devices` is a registry table referenced by `assessment_sessions` through the reported device reference. Section 9 gives the table definitions.

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4. Actors and Use Cases

This section is new in v1.1. Each use case links to the functional requirements in Section 5 that implement it.

4.1 Actors

Actor	Type	Description
Super Admin	Human, primary	Manages candidates, content, assessments, results, devices, and audit review
Candidate	Human, primary	Takes assigned assessments in the lockdown client
Technical Evaluator	Human, secondary (optional)	Manually evaluates subjective and coding answers under a rubric
Lab Staff	Human, secondary	Supervises the lab and runs the documented recovery procedure
Coding Execution Service	System	Compiles and runs candidate code and SQL; returns structured results
Cloud Monitoring	System	Raises alerts on failures and suspicious activity

4.2 Use Case Catalog

ID	Use Case	Primary Actor	Priority
UC-01	Administrator login with MFA	Super Admin	Must
UC-02	Provision candidate account	Super Admin	Must
UC-03	Bulk import candidates from CSV	Super Admin	Should
UC-04	Create and publish a question	Super Admin	Must
UC-05	Create, version, and publish an assessment	Super Admin	Must

ID	Use Case	Primary Actor	Priority
UC-06	Assign assessment to candidates or groups	Super Admin	Must
UC-07	Candidate login	Candidate	Must
UC-08	Start an assessment	Candidate	Must
UC-09	Answer and submit a question	Candidate	Must
UC-10	Handle question timeout and auto-advance	Candidate, system	Must
UC-11	Write and submit a coding solution	Candidate	Must for the full technical release
UC-12	Recover from a network or client interruption	Candidate, Lab Staff	Must
UC-13	Final submission	Candidate	Must
UC-14	Evaluate and finalize results	Super Admin, Evaluator	Must
UC-15	Correct a score with audit trail	Super Admin	Must
UC-16	Review audit log	Super Admin	Must
UC-17	Register and manage lab devices	Super Admin	Should
UC-18	Reset password or deactivate candidate	Super Admin	Must
UC-19	Export results and reports	Super Admin	Should

4.3 Use Case Specifications

UC-02: Provision Candidate Account

Actor: Super Admin. **Linked requirements:** FR-AUTH-009, FR-CAN-001, FR-CAN-007, FR-AUD-001.

Preconditions: Super Admin is authenticated with sufficient privilege.

Main flow:

1. Super Admin opens the candidate form and enters candidate ID, name, username, email, institution, qualification, graduation year, and recruitment drive.
2. The system validates required fields and checks the candidate ID and username are unique.
3. The system generates a temporary credential through the controlled provisioning workflow.
4. The system stores the password only as a secure hash and creates the profile in Active status.
5. The system writes an audit event; the temporary credential is not logged or placed in general exports.

Alternative flows: (2a) Duplicate ID or username: the system rejects the record and shows the conflict for administrator review. (3a) The candidate must change the temporary credential at first login.

Postcondition: An active candidate account exists and no plaintext password is stored.

UC-05: Create, Version, and Publish an Assessment

Actor: Super Admin. **Linked requirements:** FR-ASMT-001 to FR-ASMT-008, FR-QB-008, FR-QB-009.

Preconditions: Published questions exist in the question bank.

Main flow:

1. Super Admin enters title, role category, instructions, schedule window, overall duration, attempt limit, and marking rules.
2. Super Admin selects questions, orders them, sets per-question time limits, and configures randomization and result visibility.

3. The system validates the configuration, including that no question deadline can exceed the overall duration.
4. Super Admin publishes; the system freezes a versioned snapshot of the configuration and question versions.
5. The system records the publication in the audit log.

Alternative flows: (3a) Validation fails: the system lists errors and keeps the assessment in Draft. (4a) Edits after candidates start create a new configuration version; active attempts keep the version they began with.

Postcondition: The assessment is Scheduled or Active with an immutable version snapshot.

UC-08: Start an Assessment

Actor: Candidate. **Linked requirements:** FR-UI-001, FR-UI-002, FR-ASMT-006, FR-TIME-002, BR-002, BR-007, BR-015.

Preconditions: Candidate is logged in, the account is active, the assessment is assigned and inside its availability window, and attempts remain.

Main flow:

1. The client lists assigned assessments with eligibility and status.
2. Candidate selects the assessment and reads the instructions, including the monitoring and lockdown policy.
3. Candidate acknowledges the instructions and requests to start.
4. The server verifies eligibility, attempt count, device and client version, and that no other active session exists.
5. The server creates the session, sets the overall deadline, and activates the first question with its deadline.
6. The client displays the question and the server-provided remaining time.

Alternative flows: (4a) Another active session exists: start is denied (BR-007). (4b) Unsupported client version: the server rejects the start and instructs the candidate to contact lab staff.

Postcondition: Session is Active and the first question attempt is Active.

UC-09: Answer and Submit a Question

Actor: Candidate. **Linked requirements:** FR-UI-003 to FR-UI-007, FR-ANS-001 to FR-ANS-007, FR-TIME-009.

Preconditions: Session is Active and the question attempt is Active and before its deadline.

Main flow:

1. Candidate enters an answer; for long-form or coding questions the client autosaves drafts without extending the deadline.
2. Candidate selects Submit and Next.
3. The client sends the answer with an idempotency key.
4. The server verifies session, question identifier, status, deadline, and answer format.
5. The server stores the answer durably and then returns an acknowledgement.
6. The client shows confirmation only after the acknowledgement and displays the next question.

Alternative flows: (4a) Deadline already passed: the server rejects the answer and applies UC-10. (5a) Response lost: the client retries with the same idempotency key and the server returns the original acknowledgement without creating a duplicate.

Postcondition: The answer is stored once and the next attempt is Active, or the final-submission path begins.

UC-10: Handle Question Timeout and Auto-Advance

Actor: System (server timing service). **Linked requirements:** FR-TIME-004 to FR-TIME-007, FR-SCORE-006.

Trigger: The server-computed question deadline is reached with no accepted answer.

Main flow:

1. The server marks the question attempt Timed Out and rejects further changes.
2. The server applies the configured marking rule for unanswered questions.
3. If auto-advance is enabled, the server activates the next question with a fresh deadline.
4. On the next client sync, the client shows the timeout notice and the new question.

Alternative flows: (3a) The overall deadline is reached at the same moment: it takes precedence and the session moves to Expired (FR-TIME-007). (3b) Auto-advance is disabled: the server waits for the candidate to request the next question.

Postcondition: The timed-out attempt is final; no late answer can reopen it.

UC-11: Write and Submit a Coding Solution

Actor: Candidate; Coding Execution Service. **Linked requirements:** FR-CODE-001 to FR-CODE-014, NFR-SEC-008.

Preconditions: A coding question is Active and the submission limit is not exhausted.

Main flow:

1. Candidate selects a supported language and writes code in the editor, optionally running visible sample cases.
2. Candidate submits; the backend validates the request and queues a job.
3. The sandbox compiles and runs the code against hidden test cases under CPU, memory, process, filesystem, network, and time limits.
4. The sandbox returns a structured status: Accepted, Wrong Answer, Compilation Error, Runtime Error, Time Limit Exceeded, Memory Limit Exceeded, or Internal Evaluation Error.
5. The backend computes marks from the configured test-case weights and stores the source code and result.
6. The candidate sees only the status and permitted sample output, never hidden inputs or expected outputs.

Alternative flows: (4a) Infrastructure failure: the result is recorded as Internal Evaluation Error, not as a candidate failure, and the submission is re-queued or flagged for administrator review.

Postcondition: Source code, status, and awarded marks are stored for the attempt.

UC-12: Recover from a Network or Client Interruption

Actor: Candidate, Lab Staff. **Linked requirements:** FR-TIME-010, FR-ANS-007, FR-LAB-007, BR-010, NFR-REL-004.

Trigger: Connection loss, application crash, or lab computer restart during an Active session.

Main flow:

1. The client shows the connection status and may keep a local draft; a draft is not an accepted submission.
2. The server keeps all deadlines running and may mark the session Interrupted.
3. The candidate reconnects or restarts the client and logs in again.
4. The server returns the authoritative session state, current question, and remaining time.
5. The candidate continues; any already accepted answers stay as stored.

Alternative flows: (3a) The failure is a lab-wide or cloud outage: Lab Staff follow the incident procedure, and an authorized administrator may pause, extend, or reschedule. Every such override is logged with reason and administrator identity (BR-012).

Postcondition: The session resumes from authoritative state with no duplicate answers and no reset deadlines.

UC-14: Evaluate and Finalize Results

Actor: Super Admin, Technical Evaluator. **Linked requirements:** FR-SCORE-001 to FR-SCORE-009.

Main flow:

1. After final submission, the backend scores objective answers against the assessment version's answer key, applying negative marking.
2. Coding and SQL results are scored from approved test-case rules.
3. Pending manual items appear in an evaluation queue; evaluators award marks using the rubric and add comments.
4. While any item is pending, the result is marked Provisional.
5. When all items are complete, the system finalizes the result and computes ranking under the approved tie-break rules.

Postcondition: A Final result exists that reflects the configuration used during the attempt.

UC-15: Correct a Score with Audit Trail

Actor: Super Admin. **Linked requirements:** FR-SCORE-010, BR-012, FR-AUD-001.

Main flow:

1. Super Admin opens a result, selects the question or total to correct, and enters the revised value and a mandatory reason.
2. The system recalculates dependent totals and rankings.
3. The system records the original value, revised value, reason, timestamp, and administrator identity.

Postcondition: The corrected score is stored and the change is permanently traceable.

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5. Functional Requirements

Priorities follow the conventions in Document Control. Requirements marked **(new)** were added in v1.1; all others carry over from v1.0, restated in table form.

5.1 Authentication and Account Management

ID	Requirement	Priority
FR-AUTH-001	The system shall provide a secure Super Admin login and grant access only after server-side credential validation.	Must
FR-AUTH-002	The candidate application shall require the username and password issued by an authorized administrator. Candidates shall not self-register.	Must
FR-AUTH-003	Passwords shall be stored with an appropriate hashing algorithm such as Argon2id. Plaintext passwords shall never appear in the database or logs.	Must
FR-AUTH-004	The system shall rate-limit logins and apply temporary restriction or additional verification after repeated failures, per security policy.	Must
FR-AUTH-005	The Super Admin shall be able to start a candidate password reset through an authorized workflow. The system shall never reveal an existing password.	Must

ID	Requirement	Priority
FR-AUTH-006	The Super Admin shall be able to deactivate a candidate account. A deactivated candidate cannot start a new session; existing sessions follow the configured revocation policy.	Must
FR-AUTH-007	The backend shall maintain session state and enforce expiry, revocation, and authorization.	Must
FR-AUTH-008	Privileged administrator accounts shall support multi-factor authentication.	Must for production privileged access
FR-AUTH-009	The system shall provide a secure credential provisioning workflow. Temporary credentials require a controlled first login or authorized reset, and are never exposed in general exports or audit logs.	Must
FR-AUTH-010 (new)	A candidate session token shall be bound to the registered device and client version that created it, and shall be rejected from any other device.	Should

5.2 Candidate Management

ID	Requirement	Priority
FR-CAN-001	The Super Admin shall create candidate records with candidate ID, full name, username, email, phone (where required), institution, degree, graduation year, recruitment drive, and account status.	Must
FR-CAN-002	The Super Admin shall edit permitted candidate details; significant changes shall be logged.	Must

ID	Requirement	Priority
FR-CAN-003	The system should support bulk CSV import with a validated template; invalid rows are reported and never create incomplete records.	Should
FR-CAN-004	The Super Admin shall search and filter by candidate ID, name, assignment, account status, and recruitment drive.	Must
FR-CAN-005	The Super Admin shall assign one or more assessments to selected candidates.	Must
FR-CAN-006	The Super Admin shall view authorized login history, assignments, attempt and submission status, and results.	Must
FR-CAN-007	The system shall enforce unique candidate IDs and usernames; duplicates are handled through explicit validation and administrator review.	Must

5.3 Assessment Management

ID	Requirement	Priority
FR-ASMT-001	The Super Admin shall create an assessment with a unique identifier and configurable parameters.	Must
FR-ASMT-002	The system shall support configuring name, description, role, instructions, start and end time, overall duration, question selection and order, per-question limits, total marks, negative marking, maximum attempts, passing criteria, randomization, result visibility, auto-submission policy, and candidate eligibility.	Must
FR-ASMT-003	Assessments shall be editable before candidates begin. Once active, changes that affect existing candidates shall be restricted or versioned and shall never silently alter an active candidate's deadlines or marking rules.	Must

ID	Requirement	Priority
FR-ASMT-004	The Super Admin shall schedule assessments for a date and time range.	Must
FR-ASMT-005	The system shall support assigning assessments to individuals or defined candidate groups.	Must
FR-ASMT-006	The Super Admin shall configure the maximum number of attempts; the server shall enforce it.	Must
FR-ASMT-007	Assessments shall support Draft, Scheduled, Active, Completed, Suspended, and Archived statuses; only authorized administrators change status.	Must
FR-ASMT-008	The system should version assessment configuration so each attempt links to the exact configuration used.	Should

5.4 Question Bank Management

ID	Requirement	Priority
FR-QB-001	The Super Admin shall create questions with text, instructions, code snippets, and supported media.	Must
FR-QB-002	Questions shall be categorized by topic, skill, difficulty, role, and type.	Must
FR-QB-003	Supported types: single-answer MCQ, multiple-select, true/false, short answer, SQL, code-output prediction, debugging, programming problem, and scenario-based technical question.	Must
FR-QB-004	For objective questions the Super Admin shall create, edit, and order options. Correct-answer data shall be inaccessible to candidate-facing APIs.	Must
FR-QB-005	Each question shall support maximum marks and, where applicable, negative marks.	Must

ID	Requirement	Priority
FR-QB-006	Each question shall have an independently configurable time limit, and limits may differ by question type.	Must
FR-QB-007	The system shall validate required fields before a question can be published.	Must
FR-QB-008	Questions used in active or completed assessments shall not be modified destructively. The system should preserve question versions or snapshots.	Must
FR-QB-009	Where enabled, the system shall randomize eligible questions or options without changing marks, deadlines, or answer-key correctness.	Must
FR-QB-010	The system should support structured import and export in a documented format; exports containing answer keys are restricted to authorized administrators.	Should

5.5 Candidate Assessment Interface

ID	Requirement	Priority
FR-UI-001	After login the candidate shall see assigned assessments with eligibility and status.	Must
FR-UI-002	Instructions shall be displayed before the assessment begins and acknowledged before proceeding.	Must
FR-UI-003	In sequential mode the application shall show one active question at a time.	Must
FR-UI-004	The candidate shall not advance until the current question is submitted, unless the timer-expiry or administrative recovery policy applies.	Must
FR-UI-005	The interface shall not offer unrestricted question navigation in sequential mode.	Must

ID	Requirement	Priority
FR-UI-006	The interface shall display remaining time for the current question and, where enabled, the overall assessment.	Must
FR-UI-007	The application shall clearly show whether a question or the final assessment was submitted successfully.	Must
FR-UI-008	A candidate attempting to open another candidate's assessment or a restricted screen shall be denied.	Must
FR-UI-009	The application shall handle attempts to close or leave the assessment under the lockdown and recovery policy, and shall not claim to block every operating-system escape path.	Must

5.6 Timing and Automatic Submission

ID	Requirement	Priority
FR-TIME-001	The backend shall be authoritative for all question and overall deadlines.	Must
FR-TIME-002	The server shall record the start timestamp of each question attempt when it becomes active.	Must
FR-TIME-003	The server shall compute each question deadline from its recorded start time and configured limit.	Must
FR-TIME-004	On expiry the server shall reject further changes to the question and apply the configured submission policy.	Must
FR-TIME-005	In sequential assessments the system shall advance to the next question after a timeout when auto-advance is enabled.	Must
FR-TIME-006	The server shall enforce any overall deadline independently of question timers.	Must

ID	Requirement	Priority
FR-TIME-007	If question and overall deadlines expire together, the overall deadline takes precedence.	Must
FR-TIME-008	Changing the candidate computer's clock shall not extend or shorten any server deadline.	Must
FR-TIME-009	Repeated submissions shall not produce duplicate marks or duplicate completion records.	Must
FR-TIME-010	After a temporary network failure the backend shall return the authoritative state and remaining time. Disconnected time counts toward the deadline unless an authorized interruption policy provides otherwise.	Must
FR-TIME-011	The system shall record question start, submission, timeout, and assessment completion timestamps.	Must

5.7 Answer Submission and Storage

ID	Requirement	Priority
FR-ANS-001	The system shall accept answers only for the candidate's active, authorized question attempt.	Must
FR-ANS-002	The backend shall verify session, question identifier, status, deadline, and answer format before accepting a submission.	Must
FR-ANS-003	Accepted answers shall be stored on the server.	Must
FR-ANS-004	Submitted answers are immutable unless an explicitly authorized and audited reopening workflow permits change.	Must
FR-ANS-005	For long-form and coding questions the application should autosave drafts at a configurable interval, without extending the deadline or bypassing submission.	Should

ID	Requirement	Priority
FR-ANS-006	The backend shall return an authoritative acknowledgement after recording a submission; the client shall show success only after receiving it.	Must
FR-ANS-007	The system shall recover safely after temporary connectivity failures without duplicating accepted submissions.	Must
FR-ANS-008	On completion the system shall mark the assessment submitted and prevent unauthorized changes.	Must
FR-ANS-009 (new)	Every answer and code submission request shall carry a client-generated idempotency key; a repeated key returns the original outcome and creates no new record.	Must

5.8 Technical Assessment and Coding Engine

ID	Requirement	Priority
FR-CODE-001	The system shall support C, C++, Java, Python, and JavaScript. Exact language versions shall be defined and pinned before deployment.	Must for the full technical release
FR-CODE-002	The candidate application shall provide a suitable code editor or approved integrated editor component.	Must
FR-CODE-003	The system shall support visible sample inputs and expected outputs.	Must
FR-CODE-004	The Super Admin shall configure hidden test cases. Hidden inputs, expected outputs, and scoring details shall not be disclosed to candidates.	Must

ID	Requirement	Priority
FR-CODE-005	Candidate code shall be sent to a dedicated coding execution service for compilation or execution.	Must
FR-CODE-006	The coding engine shall enforce configurable execution time and memory limits.	Must
FR-CODE-007	Candidate code shall run in an isolated environment with restricted filesystem, network, process privileges, and resource use.	Must
FR-CODE-008	The engine shall return a status: Accepted, Wrong Answer, Compilation Error, Runtime Error, Time Limit Exceeded, Memory Limit Exceeded, or Internal Evaluation Error.	Must
FR-CODE-009	Candidate-visible responses shall not reveal hidden test cases, expected outputs, internal paths, environment variables, or infrastructure details.	Must
FR-CODE-010	The Super Admin should be able to configure the permitted number of code submissions per question.	Should
FR-CODE-011	Coding scores shall be calculated from the configured test-case and marking rules.	Must
FR-CODE-012	An infrastructure failure in the evaluation service shall be distinguished from a candidate compilation or runtime failure.	Must
FR-CODE-013	The execution service shall not receive unrestricted access to the primary database or production application credentials.	Must
FR-CODE-014	Submitted source code and evaluation results shall be retained per the configured retention policy.	Must

5.9 SQL Assessment

ID	Requirement	Priority
FR-SQL-001	The Super Admin shall author SQL questions with instructions, schema definitions, and evaluation rules.	Must
FR-SQL-002	SQL submissions shall execute only against an isolated, controlled assessment database or sandbox.	Must
FR-SQL-003	The evaluation environment shall block access to production data, external databases, operating-system resources, and other candidates' submissions.	Must
FR-SQL-004	The system should compare query results with approved expected results or run predefined validation tests.	Should
FR-SQL-005	The SQL evaluation service shall enforce query execution limits and restrict expensive or unsafe operations.	Must

5.10 Scoring and Evaluation

ID	Requirement	Priority
FR-SCORE-001	The backend shall evaluate objective questions against the answer key of the assessment version.	Must
FR-SCORE-002	The system shall support configurable negative marking.	Must
FR-SCORE-003	The system should support partial credit for multiple-select questions, coding test cases, and manually evaluated responses.	Should
FR-SCORE-004	Short-answer questions shall support manual evaluation unless a separately approved mechanism is implemented.	Must

ID	Requirement	Priority
FR-SCORE-005	Marks shall be calculated according to the configured assessment and question rules.	Must
FR-SCORE-006	The configured marking rule shall be applied to unanswered and timed-out questions.	Must
FR-SCORE-007	Candidates shall not be able to modify scores, answer keys, marks, or evaluation status through client requests.	Must
FR-SCORE-008	Final results shall be generated after all required evaluation completes; while manual evaluation is pending the result is marked provisional.	Must
FR-SCORE-009	The system should support ranking by approved score and tie-breaking rules.	Should
FR-SCORE-010	Score corrections shall create an audit record with original value, revised value, reason, timestamp, and administrator identity.	Must

5.11 Super Admin Dashboard and Reporting

ID	Requirement	Priority
FR-ADMIN-001	The dashboard shall give an overview of candidates, assessments, active sessions, submissions, and results.	Must
FR-ADMIN-002	The Super Admin shall monitor assessment status and identify candidates with interrupted sessions.	Must
FR-ADMIN-003	The dashboard shall show authorized candidate results and assessment-level statistics.	Must
FR-ADMIN-004	The system should export authorized results to CSV or Excel.	Should
FR-ADMIN-005	The system should generate reports with identity, assessment details, completion status, score, and permitted evaluation details.	Should

ID	Requirement	Priority
FR-ADMIN-006	The system should provide question-level statistics, including response distribution and correct-answer rates.	Should
FR-ADMIN-007	The Super Admin shall search and filter audit events.	Must
FR-ADMIN-008	Exports containing candidate data or answer keys shall be restricted to authorized users and logged.	Must
FR-ADMIN-009	Reports shall be generated from authoritative server-side records.	Must

5.12 Lab Client and Kiosk Management

ID	Requirement	Priority
FR-LAB-001	The system should maintain a registry of authorized lab computers.	Should
FR-LAB-002	Each registered device shall have a unique application-managed identifier; security decisions shall not rely solely on a client-provided identifier.	Must
FR-LAB-003	The backend shall identify the installed client version reported by the application.	Must
FR-LAB-004	The system should reject or restrict unsupported client versions.	Should
FR-LAB-005	The deployment shall integrate the candidate application with the selected OS kiosk or lockdown configuration.	Must
FR-LAB-006	The system shall record detected application exits, focus changes, or lockdown events where technically available.	Must
FR-LAB-007	Authorized lab staff shall have a documented recovery procedure for legitimate technical failures.	Must

ID	Requirement	Priority
FR-LAB-008	The implementation shall not rely solely on full-screen mode or client-side JavaScript to prevent operating-system switching.	Must

5.13 Audit Logging

ID	Requirement	Priority
FR-AUD-001	The system shall record significant administrative actions: candidate creation, assessment publication, assignment changes, question changes, account deactivation, and result correction.	Must
FR-AUD-002	The system shall record session start, question activation, submission, timeout, interruption, and final completion.	Must
FR-AUD-003	The system shall record login successes, relevant failures, session revocation, and administrative authentication events.	Must
FR-AUD-004	Audit records shall be protected against unauthorized modification and deletion.	Must
FR-AUD-005	Audit entries should include event type, timestamp, actor, affected record, outcome, and a request or correlation identifier.	Should
FR-AUD-006	Passwords, tokens, secret keys, and unnecessary sensitive candidate data shall not be logged.	Must
FR-AUD-007	Audit logs shall be retained per an approved retention policy.	Must

5.14 System Configuration

ID	Requirement	Priority
FR-CONF-001	The Super Admin shall manage approved assessment and system settings through authorized controls.	Must
FR-CONF-002	The backend shall validate configuration values and reject invalid or unsupported settings.	Must
FR-CONF-003	Security-sensitive configuration changes shall be logged.	Must
FR-CONF-004	Development, testing, and production shall use separate configurations and credentials.	Must

6. Business Rules

ID	Rule
BR-001	Only authorized administrators may create or activate candidate accounts.
BR-002	A candidate may access only assessments assigned to that candidate or an authorized candidate group.
BR-003	In sequential mode the candidate must submit the current question before proceeding.
BR-004	The backend determines assessment state, deadlines, submission acceptance, and final scores.
BR-005	An expired question accepts no modifications; the timeout and auto-advance policy decide the next action.
BR-006	An overall deadline takes precedence over a question deadline that expires at the same time.

ID	Rule
BR-007	By default one active attempt is allowed per candidate per assignment; extra simultaneous sessions are denied unless an approved policy permits them.
BR-008	A completed assessment is not editable by the candidate.
BR-009	Candidate-facing interfaces never receive correct answers or hidden test data, unless an explicit post-assessment disclosure policy permits it.
BR-010	Network or client interruptions never reset deadlines or grant extra attempts.
BR-011	Scores are calculated from the configuration associated with the candidate's attempt.
BR-012	Any authorized reopening, deadline extension, or score change is logged with reason and administrator identity.
BR-013	Candidate code never runs in the main application process or with unrestricted access to production infrastructure.
BR-014	Only candidate information required for recruitment is collected and retained.
BR-015	Candidates are told about monitoring, lockdown, and integrity policies before starting.

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7. Non-Functional Requirements

7.1 Performance

ID	Requirement
NFR-PERF-001	Under the approved target load and normal network conditions, 95% of login requests should complete within 2 seconds.
NFR-PERF-002	95% of question-delivery requests should complete within 1.5 seconds.
NFR-PERF-003	95% of ordinary answer submissions should receive a server acknowledgement within 2 seconds. Coding execution has separate limits.
NFR-PERF-004	The system shall be load-tested for the expected peak of concurrent candidates; the target shall be agreed before production deployment.
NFR-PERF-005	Reports should be generated within an agreed time for the expected volume; large exports may run asynchronously.

7.2 Security

ID	Requirement
NFR-SEC-001	All client-to-backend communication shall use HTTPS with correctly validated TLS certificates.
NFR-SEC-002	Passwords shall be stored using an appropriate salted password-hashing scheme.
NFR-SEC-003	Every component and user shall receive only the permissions its function requires.
NFR-SEC-004	The backend shall enforce authorization on every protected operation.

ID	Requirement
NFR-SEC-005	Database passwords, cloud credentials, signing keys, and other secrets shall live in an approved secrets-management mechanism, never in source code or client binaries.
NFR-SEC-006	The system shall use parameterized database operations and validated inputs against injection.
NFR-SEC-007	Sessions and tokens shall have secure handling, appropriate expiry, revocation, and replay protection.
NFR-SEC-008	The coding service shall enforce process, memory, CPU, filesystem, network, and time restrictions.
NFR-SEC-009	Audit logs shall be protected from unauthorized alteration.
NFR-SEC-010	The application, OS, runtime, dependencies, and infrastructure shall follow a documented patch and vulnerability-management process.
NFR-SEC-011	Dependency scanning, authentication and authorization testing, and penetration testing appropriate to the risk shall be completed before the first live recruitment drive.
NFR-SEC-012	The web application and APIs shall be designed and tested against applicable OWASP guidance.

7.3 Reliability and Availability

ID	Requirement
NFR-REL-001	Proposed initial target: 99.5% monthly availability for the production assessment service, excluding agreed maintenance windows, reviewed against budget and business needs.

ID	Requirement
NFR-REL-002	Accepted submissions shall be committed to durable server storage before success is acknowledged.
NFR-REL-003	Retries shall not create duplicate accepted submissions or duplicate score records.
NFR-REL-004	The system shall recover from application restarts and temporary network interruptions without resetting authoritative deadlines.
NFR-REL-005	Errors shall be recorded with correlation identifiers; user-facing messages shall not expose internal details.
NFR-REL-006	The database shall be backed up regularly and restoration tested before production approval.

7.4 Usability, Maintainability, Scalability, Privacy, Compatibility

ID	Requirement
NFR-USE-001	The candidate interface shall give clear instructions and minimize unnecessary navigation.
NFR-USE-002	The current question timer shall remain clearly visible.
NFR-USE-003	The application shall clearly communicate submission, timeout, reconnection, and final completion states.
NFR-USE-004	Common administrative operations shall sit in a structured dashboard with validation and confirmation for destructive actions.
NFR-USE-005	The system should support keyboard accessibility, readable text, sufficient contrast, and assistive technologies where compatible with the lockdown policy.

ID	Requirement
NFR-MNT-001	Authentication, assessment management, timing, submissions, evaluation, reporting, and code execution shall be separate modules.
NFR-MNT-002	The codebase shall follow consistent standards, meaningful names, appropriate error handling, and automated tests.
NFR-MNT-003	The project shall include architecture, database, API, deployment, and troubleshooting documentation.
NFR-MNT-004	Application code and infrastructure configuration shall be kept in version control.
NFR-MNT-005	Schema changes shall use controlled, versioned migrations.
NFR-SCALE-001	Stateless API services shall support horizontal scaling where required.
NFR-SCALE-002	The database and the coding execution service shall be sized and scaled independently.
NFR-SCALE-003	The application shall be load-tested with realistic assessment and coding workloads before candidate capacity is increased.
NFR-PRIV-001	Candidate information shall be collected only for approved recruitment and assessment purposes.
NFR-PRIV-002	Access to candidate information shall be restricted by role and business need.
NFR-PRIV-003	The organization shall define retention, deletion, access, and disclosure policies.
NFR-PRIV-004	Monitoring and lockdown activity shall be described in candidate instructions and handled under institutional policy and applicable law.
NFR-COMP-001	The candidate application shall be tested against the approved Windows version, device configuration, resolution, and kiosk policy.

ID	Requirement
NFR-COMP-002	The administrator dashboard shall support approved versions of common desktop browsers.
NFR-COMP-003	The system shall validate client versions and reject unsupported ones according to deployment policy.

7.5 Proposed Recovery Objectives

Measure	Proposed Target
Recovery Point Objective (RPO)	15 minutes or less
Recovery Time Objective (RTO)	4 hours or less
Database backup	Daily full backup plus point-in-time recovery where available
Restore testing	Before production release and periodically thereafter
Submission durability	Every acknowledged submission is durably recorded

These are design targets for approval, not guarantees. The cloud provider, database configuration, service tier, and backup strategy must support them.

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8. External Interface Requirements

8.1 Admin Web Interface

The dashboard shall provide login and logout; candidate, assessment, and question-bank management; scheduling and assignment; assessment monitoring; results and reports; audit-log search; and configuration and account management. It shall show clear validation errors and confirmation prompts for destructive or security-sensitive actions.

8.2 Candidate Desktop Interface

The desktop application shall provide login, eligibility and instructions, assessment start, current question display, question timer, answer entry, a code editor for programming questions, Submit and Next, timeout notification, reconnection handling, and final submission confirmation. It shall not display administrative controls or unrestricted navigation.

8.3 Backend API

Clients use versioned APIs over HTTPS. Every protected endpoint authenticates and authorizes the caller, validates request data, enforces assessment state, and returns structured responses. Endpoint names are illustrative and may change during implementation.

Group	Method and Path	Purpose
Auth	POST /api/v1/auth/login	Sign in
Auth	POST /api/v1/auth/logout	End session
Auth	POST /api/v1/auth/refresh	Refresh token, if that architecture is chosen
Auth	POST /api/v1/auth/password-reset	Reset flow
Candidates	GET, POST /api/v1/admin/candidates	List, create
Candidates	GET, PATCH /api/v1/admin/candidates/{id}	Read, update
Candidates	POST /api/v1/admin/candidates/import	Bulk CSV import
Assessments	GET, POST /api/v1/admin/assessments	List, create
Assessments	GET, PATCH /api/v1/admin/assessments/{id}	Read, update
Assessments	POST /api/v1/admin/assessments/{id}/publish	Publish a version
Assessments	POST /api/v1/admin/assessments/{id}/assignments	Assign candidates or groups
Candidate	GET /api/v1/candidate/assessments	Assigned assessments
Candidate	POST /api/v1/candidate/assessments/{id}/start	Start a session

Group	Method and Path	Purpose
Candidate	GET /api/v1/candidate/sessions/{id}/current-question	Active question and deadline
Candidate	POST /api/v1/candidate/sessions/{id}/answers	Submit an answer
Candidate	POST /api/v1/candidate/sessions/{id}/code-submissions	Submit code
Candidate	POST /api/v1/candidate/sessions/{id}/final-submit	Final submission
Candidate	GET /api/v1/candidate/sessions/{id}/status	Authoritative state and remaining time
Results	GET /api/v1/admin/results, /results/{id}	Results
Audit	GET /api/v1/admin/audit-logs	Audit search

8.3.1 Illustrative Payloads (new in v1.1)

These examples show intent only; field names may change. Note that the client never sends a deadline, and the server never sends an answer key.

Submit an answer (request):

```
POST /api/v1/candidate/sessions/9f3c/answers
Idempotency-Key: 6b1e2d0a-4c8f-4e0b-9a52-0d7d3f1c8a11

{
  "questionAttemptId": "qa_0142",
  "answer": { "type": "single_choice", "optionId": "opt_3" }
}
```

Accepted (response):

```
{
  "status": "accepted",
  "questionAttemptId": "qa_0142",
  "submittedAt": "2026-11-02T09:41:07Z",
  "next": { "questionAttemptId": "qa_0143", "deadlineAt": "2026-11-
02T09:43:07Z" },
  "serverTime": "2026-11-02T09:41:07Z"
}
```

Rejected after the deadline (response):

```
{
  "status": "rejected",
  "code": "QUESTION_EXPIRED",
  "message": "This question has timed out.",
  "correlationId": "c-7d21f8"
}
```

8.3.2 API Security Requirements

1. Candidate APIs return only the candidate's authorized assessment data.
2. Candidate requests never determine the authoritative deadline.
3. Administrative endpoints require appropriate privileges.
4. Answer keys and hidden test cases are never returned through candidate endpoints.
5. Requests are validated against the current assessment state.
6. Sensitive operations are logged.
7. Rate limits apply to authentication and resource-intensive operations.
8. Idempotency or equivalent duplicate protection applies where necessary (FR-ANS-009).
9. Errors never disclose secrets or internal infrastructure details.
10. The API enforces request-size and resource-consumption limits.

8.3.3 Standard Error Codes (new in v1.1)

Code	HTTP	Meaning
UNAUTHENTICATED	401	Missing or invalid session
FORBIDDEN	403	Role or ownership check failed

Code	HTTP	Meaning
NOT_ELIGIBLE	403	Outside window, attempts used, or assignment revoked
INVALID_STATE	409	Transition not allowed for the current session or question state
QUESTION_EXPIRED	409	Deadline passed before the answer arrived
DUPLICATE_REQUEST	200	Idempotency key already processed; original result returned
UNSUPPORTED_CLIENT	426	Client version is not approved
RATE_LIMITED	429	Too many requests
VALIDATION_FAILED	422	Request body failed validation
INTERNAL_ERROR	500	Unexpected failure; correlation ID returned

8.4 Database Interface

The backend reaches the relational database only through a controlled data-access layer. Candidate clients never connect to it. Database access uses least-privilege service credentials, encrypted connections where applicable, and restricted network access.

8.5 Coding Execution Interface

The backend calls the coding service over a secured internal interface that carries only the source code, language configuration, test-execution parameters, and authorized evaluation identifiers. The service returns a structured result without internal infrastructure details or hidden test data.

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9. Database Requirements and Logical Data Model

The database keeps normalized records for identity, candidate profiles, assessments, questions, attempts, answers, coding submissions, results, and audit events. The physical schema is confirmed during database design. Relationships are shown in Section 3.5.

Fields marked **(new)** were added in v1.1.

9.1 Proposed Tables

Table	Purpose	Key Fields
users	Login identity and access control	user_id (PK), username (unique), password_hash, role, account_status (active, inactive, locked), created_at, updated_at
candidate_profiles	Recruitment information	candidate_id (PK), user_id (FK), full_name, email, phone, institution, qualification, graduation_year, recruitment_reference
assessments	Assessment configuration	assessment_id (PK), title, description, role_category, overall_duration, start_at, end_at, status, attempt_limit, configuration_version, created_by
questions	Question content and metadata	question_id (PK), question_type, question_text, topic, difficulty, maximum_marks, negative_marks, question_version, status
question_options	Options for objective questions	option_id (PK), question_id (FK), option_text, display_order. Correct-answer data is held in a protected server-side structure inaccessible to candidate APIs
assessment_questions	Questions mapped to an assessment	assessment_question_id (PK), assessment_id, question_id, sequence_number, time_limit_seconds, marks_override, question_version

Table	Purpose	Key Fields
assessment_assignments	Candidate assignments	assignment_id (PK), assessment_id, candidate_id, assignment_status (assigned, started, completed, revoked), assigned_at, assigned_by
assessment_sessions	Attempt state	session_id (PK), assignment_id, attempt_number, started_at, overall_deadline, session_status (active, interrupted, submitted, expired), submitted_at, client_version, device_id **(new)**
question_attempts	State and timing per question	question_attempt_id (PK), session_id, assessment_question_id, started_at, deadline_at, submitted_at, attempt_status (active, submitted, timed out, skipped), awarded_marks
answers	Submitted responses	answer_id (PK), question_attempt_id, answer_payload, submitted_at, evaluation_status (pending, evaluated, failed), awarded_marks, idempotency_key **(new, unique per session)**
coding_submissions	Code and evaluation reference	coding_submission_id (PK), question_attempt_id, language, source_code or protected object reference, submission_status (queued, running, completed, failed), execution_reference, awarded_marks

Table	Purpose	Key Fields
test_cases	Coding test cases	test_case_id (PK), question_id, input_data, expected_output, visibility (sample or hidden), marks. Hidden cases are readable only by evaluation services and administrators
results	Finalized results	result_id (PK), session_id, total_score, maximum_score, result_status (provisional or final), finalized_at
audit_logs	Security and administrative history	audit_id (PK), actor_user_id, event_type, affected_entity, event_timestamp, outcome (success, denied, failure), correlation_id, metadata (validated, non-sensitive)
lab_devices	Authorized candidate computers	device_id (PK), device_reference (application-managed unique identifier), device_status (active, inactive, blocked), approved_client_version, last_seen_at
score_corrections **(new)**	History of authorized score changes	correction_id (PK), result_id, original_value, revised_value, reason, corrected_by, corrected_at

9.2 Data Integrity

The database shall enforce primary-key uniqueness, foreign-key relationships, required fields, unique usernames and candidate identifiers, valid status transitions for assessments and sessions, referential integrity for assignments and attempts, transactional persistence of accepted submissions, and indexes for candidate search, active sessions, and reports.

9.3 Data Retention

The organization defines retention periods before production deployment. The system shall support controlled deletion, archival, or anonymization where policy or law requires it, taking audit requirements and legal preservation obligations into account.

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10. Security Architecture and Threat Model

10.1 Security Objectives

The design protects candidate credentials and personal data, unpublished questions and answer keys, hidden coding test cases, timing and submission state, answers and source code, final scores and reports, cloud infrastructure and service credentials, and audit logs and configuration.

10.2 Threats and Mitigations

Threat	Potential Impact	Required Mitigation
Credential guessing	Unauthorized candidate access	Rate limiting, password hashing, account controls
Candidate accessing another account	Data disclosure or impersonation	Server-side authorization and session binding
Client-side timer manipulation	Unfair extra time	Server-authoritative deadlines
Direct API manipulation	Skipped questions or unauthorized submission	State validation and authorization
Answer-key extraction	Assessment compromise	Keep answer keys server-side
Hidden test disclosure	Coding assessment compromise	Isolated evaluation and restricted responses
Code execution attack	Server compromise	Isolated sandbox with restricted privileges
SQL injection	Data theft or modification	Parameterized queries and input validation

Threat	Potential Impact	Required Mitigation
Stolen administrator credentials	Platform compromise	MFA, least privilege, monitoring
Network interception	Credential or data exposure	HTTPS and certificate validation
Duplicate submissions	Incorrect results	Idempotency keys and transactions
Candidate application exit	Interruption or cheating	Managed kiosk policies and audit events
Cloud service failure	Interrupted examinations	Monitoring, recovery procedures, backups
Unauthorized report export	Candidate information disclosure	Role checks, restricted exports, audit logs
Malicious dependency	Supply-chain compromise	Dependency pinning, scanning, controlled updates

10.3 Candidate Application Lockdown

The installed application runs full-screen under an approved lockdown configuration. Application-level controls cannot guarantee that a candidate cannot use a Windows shortcut, end a process, open another application, or use a separate device. The deployment therefore combines the application with operating-system controls: a dedicated candidate account and a tested kiosk or device-management policy.

The implementation shall:

- Restrict access to unapproved applications where supported.
- Disable or restrict unauthorized navigation and system access using supported OS policies.
- Prevent candidates from modifying assessment files or application configuration.
- Protect application updates and executable integrity.
- Provide a documented administrative recovery mechanism.
- Test keyboard shortcuts, application switching, restart behaviour, and kiosk escape paths.

- Record detected incidents without treating every focus change as conclusive proof of cheating.

10.4 Coding Sandbox Security

The sandbox is isolated from the main infrastructure. It shall run code with unprivileged identities; restrict CPU, memory, process count, disk use, and duration; restrict filesystem access; disable unnecessary outbound and lateral network access; block cloud metadata endpoints and application secrets; use controlled language runtimes and compiler versions; clean up temporary environments; limit the size of source files, inputs, outputs, and logs; distinguish candidate errors from infrastructure failures; and never run candidate code in the backend API process.

10.5 Pre-Release Security Testing

Before production release the team shall test authentication and password reset, role and permission boundaries, candidate access to other candidates' records, question and answer-key exposure, timer and submission manipulation, API replay and duplicate submissions, SQL injection and input validation, sandbox escape attempts, excessive resource consumption, kiosk restrictions and OS escape paths, and backup restoration and audit integrity.

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11. Operational Workflows

11.1 Assessment Lifecycle

Draft, Validated, Published, Scheduled or Available, Active Attempt, Submitted, Evaluated, Finalized, Archived. Not every assessment passes through every optional status. Session and question-attempt states are in Sections 3.3 and 3.4.

11.2 Overall Assessment Timing

When an overall duration is configured, the backend computes the overall deadline when the assessment begins. A question deadline never exceeds the overall deadline. The server prevents extra time through application refresh, system clock changes, or reconnection.

11.3 Network Interruption Workflow

1. The application shows a clear connection status.
2. The application may preserve a local draft; a draft is never an accepted submission.
3. The server deadline continues to apply.
4. On reconnection the client fetches the authoritative session state.
5. Already accepted submissions are not duplicated.
6. Expired questions do not reopen automatically.
7. Any administrator-authorized extension or reopening is logged.

The assessment policy shall define what happens during a prolonged cloud outage: pausing through an authorized incident procedure, cancelling affected attempts, or rescheduling candidates.

11.4 Incident Handling Roles (new in v1.1)

Situation	First Responder	Decision Maker	Required Record
One candidate's computer fails	Lab Staff	Lab Staff, with Super Admin approval for any time adjustment	Audit event and reason
Lab-wide network loss	Lab Staff	Super Admin	Incident note, affected sessions, any extension
Cloud service outage	Super Admin	Placement Coordinator	Incident timeline, pause or reschedule decision
Suspected integrity event	Lab Staff	Placement Coordinator	Audit events; no automatic penalty from a single focus-change event

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12. Cloud Deployment and Infrastructure

12.1 Deployment Architecture

Production includes publicly reachable HTTPS endpoints for approved clients, private application services where supported, restricted database access, managed secret storage, centralized logging and monitoring, protected backups, an isolated coding execution environment, and separate development, test, and production environments. The cloud provider is chosen on organizational policy, cost, available expertise, regional requirements, and service compatibility.

12.2 Environment Separation

Development, staging, and production use separate credentials and secrets, databases or isolated instances, storage locations, API configurations, and access permissions. Production candidate data is never copied to development without approved protection and handling procedures.

12.3 Release Pipeline

1. Source-code review.
2. Automated unit and integration tests.
3. Dependency and vulnerability scanning.
4. Database migration validation.
5. Staging deployment.
6. Regression testing.
7. Approval for production release.
8. Production deployment.
9. Health checks and monitoring.
10. Rollback readiness.

12.4 Monitoring and Alerting

The system monitors API availability and latency, authentication failures, active sessions, submission failures, database health, coding queue length and failures, storage capacity, resource use, backup success, and security-relevant events. Alerts cover critical service failures and suspicious activity.

12.5 Backup and Restoration

Automated backups and a documented restoration process are required. Backups are protected from unauthorized access and deletion. Restoration testing verifies that assignments, accepted answers, assessment state, and results recover to the agreed recovery point.

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13. Testing Strategy

13.1 Test Levels

Level	Scope
Unit	Authentication and permission checks, timer calculations, state transitions, question validation, scoring, negative marking, auto-submission, duplicate prevention, coding result interpretation
Integration	Dashboard and candidate client to backend, backend to database, timing and submission, coding service, result generation, audit events, recovery after interruption
System	Complete candidate and administrator workflows from account creation through finalization and reporting
Security	Authentication and sessions, authorization bypass, API manipulation, sensitive data exposure, SQL injection, rate limiting, sandbox isolation, kiosk restrictions, audit integrity

Level	Scope
Performance	Simultaneous logins, question delivery, concurrent submissions, mass timer expiry, concurrent coding submissions, final submission, result generation, report exports
Recovery	Application closure, lab computer restart, temporary network loss, backend restart, coding outage, database restoration, expired-session recovery, retry after an ambiguous submission response
User acceptance	The placement team verifies the recruitment workflow, assessment rules, and candidate experience against Section 15

13.2 Sample Test Cases (new in v1.1)

ID	Scenario	Steps	Expected Result	Requirement
TC-01	Clock manipulation	Start a question, move the lab computer's clock forward or back, then submit	The server deadline is unchanged; acceptance depends only on server time	FR-TIME-008
TC-02	Skip a question by API	Call the answers endpoint for a future question attempt	Rejected with INVALID_STATE; no record created	FR-ANS-001, BR-003
TC-03	Late submission	Wait past the deadline and submit	Rejected with QUESTION_EXPIRED; attempt is Timed Out	FR-TIME-004

ID	Scenario	Steps	Expected Result	Requirement
TC-04	Duplicate submit	Send the same answer twice with one idempotency key	One stored answer, one score; both calls return the same outcome	FR-ANS-009, FR-TIME-009
TC-05	Lost acknowledgement	Drop the response, retry after reconnecting	Original acknowledgement returned; no duplicate	FR-ANS-007
TC-06	Simultaneous deadlines	Configure a question to end exactly at the overall deadline	Session becomes Expired; overall deadline wins	FR-TIME-007
TC-07	Disconnect during question	Cut the network for part of a question, reconnect	Remaining time reflects the disconnected period; deadline not reset	FR-TIME-010, BR-010
TC-08	Cross-candidate access	Candidate A requests Candidate B's session	403 FORBIDDEN; audit event recorded	FR-UI-008, NFR-SEC-004
TC-09	Answer-key exposure	Inspect every candidate API response for a published MCQ	No correct-answer field anywhere	BR-009, FR-QB-004
TC-10	Hidden test leak	Submit failing code and inspect the response	Only status and permitted output; no hidden input or expected output	FR-CODE-009

ID	Scenario	Steps	Expected Result	Requirement
TC-11	Sandbox escape	Run code that opens network sockets, reads environment variables, or forks repeatedly	Blocked or terminated; no secret or network access	FR-CODE-007, NFR-SEC-008
TC-12	Infrastructure failure	Stop the coding service during evaluation	Status Internal Evaluation Error; candidate is not penalized	FR-CODE-012
TC-13	Negative marking	Submit a wrong answer on a question with negative marks	Score reduced by the configured value	FR-SCORE-002
TC-14	Score correction	Correct a score as Super Admin	Audit record holds original, revised, reason, time, admin	FR-SCORE-010
TC-15	Second session	Log in on a second computer while a session is Active	Start denied	BR-007
TC-16	Unsupported client	Start with a retired client version	UNSUPPORTED_CLIENT; start refused	FR-LAB-004
TC-17	Brute-force login	Submit repeated wrong passwords	Rate limiting or temporary lockout engages; events logged	FR-AUTH-004
TC-18	Kiosk escape	Try Alt+Tab, Task Manager, and restart during an assessment	Blocked by OS policy where configured; remaining paths documented and audited	FR-LAB-005 to FR-LAB-008

ID	Scenario	Steps	Expected Result	Requirement
TC-19	Backup restore	Restore the latest backup into staging	Assignments, accepted answers, and results are recovered to the recovery point	NFR-REL-006
TC-20	Mass timer expiry	Expire many question timers at the same moment under load	All advance correctly within performance targets	NFR-PERF-004

14. Risks, Dependencies, and Open Decisions

14.1 Project Risks

Risk	Impact	Mitigation
Unsupported or unpatched Windows installation	Higher security exposure	Verify update coverage or upgrade
Incomplete kiosk lockdown	Candidates reach unapproved applications	Test OS-level controls and escape paths
Internet outage	Assessment interruption	Define recovery and incident policies
Coding sandbox vulnerability	Possible infrastructure compromise	Isolated execution and security testing
Poor timer implementation	Unfair timing	Enforce deadlines on the server
Hidden test exposure	Integrity compromise	Keep test data in restricted evaluation services
Incorrect score calculation	Recruitment decisions affected	Automated scoring tests and audited corrections
Excessive cloud cost	Budget overrun	Load testing, resource limits, cost monitoring

Risk	Impact	Mitigation
Insufficient test capacity	Slow or unavailable assessments	Set a concurrency target and load-test it
Data exposure	Privacy and reputational harm	Least privilege, encryption, logging, retention controls

14.2 Dependencies

Approved cloud infrastructure; a supported backend runtime and relational database; a Windows desktop application framework; a secure coding execution solution; an approved kiosk configuration; valid TLS certificates; secure credential and secret management; institutional approval for candidate data handling; and defined assessment and marking policies.

14.3 Open Decisions

No.	Decision	Needed Before
1	Expected maximum number of concurrent candidates	Capacity sizing and load tests
2	Cloud provider and hosting region	Infrastructure setup
3	Exact Windows editions and update status in the lab	Client build and kiosk design
4	Whether candidates may resume after a technical interruption	Recovery policy
5	Assessment and question-level time limits	Content configuration
6	Coding language versions and execution limits	Sandbox build
7	Candidate data and audit-log retention periods	Production deployment
8	Whether candidates can see results	Results module
9	Whether manual evaluation is needed in the first release	Evaluator role scope

No.	Decision	Needed Before
10	Whether MFA and administrator network restrictions are mandatory	Admin hardening
11	Whether integration with existing placement or student systems is needed	Scope confirmation
12	Whether offline assessment is needed in a future release	Architecture review

These shall be documented and approved before production deployment.

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15. Acceptance Criteria

The system is production-ready only when the mandatory criteria below are met.

15.1 Authentication and Candidate Management

- [ ] Only authorized administrators can create candidate accounts.
- [ ] Candidates can log in with provisioned credentials.
- [ ] Candidate accounts can be deactivated.
- [ ] Passwords are not stored in plaintext.
- [ ] Unauthorized users cannot reach administrative endpoints.
- [ ] Candidate information is protected from other candidates.

15.2 Assessment Management

- [ ] The Super Admin can create and publish assessments.
- [ ] The Super Admin can create and manage question banks.
- [ ] Each question supports a configurable time limit.
- [ ] Assessments can be assigned to specific candidates.
- [ ] Configuration is versioned or otherwise protected against unintended changes to active attempts.

15.3 Timing and Submission

- [ ] The server controls question deadlines.
- [ ] Changing the candidate computer's clock does not alter deadlines.
- [ ] In sequential mode the candidate cannot advance before submitting the current question.
- [ ] Expired questions reject late modifications.
- [ ] Automatic advancement follows policy.
- [ ] Overall deadlines are enforced when enabled.
- [ ] Duplicate submissions create no duplicate scores.
- [ ] Final submission prevents candidate edits.

15.4 Coding and SQL

- [ ] Approved programming languages can be selected.
- [ ] Coding submissions execute in an isolated environment.
- [ ] Hidden test cases are not exposed.
- [ ] Time and memory limits are enforced.
- [ ] Compilation and runtime errors are reported appropriately.
- [ ] SQL assessments use an isolated environment.
- [ ] Infrastructure failures are distinguished from candidate errors.

15.5 Security and Lockdown

- [ ] HTTPS is enforced for client-server communication.
- [ ] Backend authorization applies to protected APIs.
- [ ] No secrets are embedded in candidate-facing code.
- [ ] Kiosk configuration is tested on the actual lab hardware.
- [ ] OS escape paths are documented and tested.
- [ ] The application does not rely solely on full-screen mode.
- [ ] Security logs capture relevant administrative and assessment events.
- [ ] Critical security findings are resolved or formally accepted.

15.6 Results and Reporting

- [ ] Objective answers are scored correctly.
- [ ] Negative marking follows configuration.
- [ ] Coding scores follow the approved test-case rules.
- [ ] Final results reflect the configuration used during the attempt.
- [ ] Authorized administrators can review and export results.
- [ ] Score corrections generate audit records.

15.7 Reliability and Operations

- [ ] Accepted answers survive application restart or reconnection.
- [ ] Deadlines are not reset by disconnection.
- [ ] Backups restore successfully.
- [ ] Performance targets are verified at the approved concurrency level.
- [ ] Monitoring and incident-response procedures are operational.
- [ ] User, deployment, and maintenance documentation is complete.

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16. Requirements Traceability Matrix

16.1 By Requirement Area

Requirement Area	Requirement IDs	Verification Method
Authentication	FR-AUTH-001 to FR-AUTH-010	Functional and security tests
Candidate management	FR-CAN-001 to FR-CAN-007	Functional tests
Assessment management	FR-ASMT-001 to FR-ASMT-008	Functional and integration tests
Question bank	FR-QB-001 to FR-QB-010	Functional tests
Candidate interface	FR-UI-001 to FR-UI-009	UI and workflow tests
Timing	FR-TIME-001 to FR-TIME-011	Timer and boundary tests

Requirement Area	Requirement IDs	Verification Method
Answer submission	FR-ANS-001 to FR-ANS-009	Integration and recovery tests
Coding assessment	FR-CODE-001 to FR-CODE-014	Functional and sandbox security tests
SQL assessment	FR-SQL-001 to FR-SQL-005	Functional and isolation tests
Evaluation	FR-SCORE-001 to FR-SCORE-010	Scoring and regression tests
Administration and reports	FR-ADMIN-001 to FR-ADMIN-009	Functional and permission tests
Lab client	FR-LAB-001 to FR-LAB-008	Device and kiosk testing
Audit logging	FR-AUD-001 to FR-AUD-007	Audit and security tests
Configuration	FR-CONF-001 to FR-CONF-004	Functional and review
Performance	NFR-PERF-001 to NFR-PERF-005	Load testing
Security	NFR-SEC-001 to NFR-SEC-012	Security testing
Reliability	NFR-REL-001 to NFR-REL-006	Recovery and availability testing
Usability	NFR-USE-001 to NFR-USE-005	User acceptance testing
Maintainability	NFR-MNT-001 to NFR-MNT-005	Code and documentation review
Scalability	NFR-SCALE-001 to NFR-SCALE-003	Architecture review and load testing
Privacy	NFR-PRIV-001 to NFR-PRIV-004	Policy and access-control review
Compatibility	NFR-COMP-001 to NFR-COMP-003	Environment compatibility testing

16.2 Use Cases to Requirements (new in v1.1)

Use Case	Main Requirements	Test Cases
UC-02 Provision candidate	FR-AUTH-009, FR-CAN-001, FR-CAN-007	Functional
UC-05 Publish assessment	FR-ASMT-001 to FR-ASMT-008	Functional
UC-08 Start assessment	FR-UI-001, FR-UI-002, FR-ASMT-006, BR-007	TC-15, TC-16
UC-09 Submit answer	FR-ANS-001 to FR-ANS-009, FR-UI-004	TC-02, TC-04, TC-05
UC-10 Timeout	FR-TIME-004 to FR-TIME-007	TC-03, TC-06
UC-11 Coding submission	FR-CODE-001 to FR-CODE-014	TC-10, TC-11, TC-12
UC-12 Recovery	FR-TIME-010, BR-010, NFR-REL-004	TC-07, TC-19
UC-14 Finalize results	FR-SCORE-001 to FR-SCORE-009	TC-13
UC-15 Score correction	FR-SCORE-010, BR-012	TC-14

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17. Recommended Implementation Plan

Implement in controlled phases.

Phase	Focus	Deliverables
1. Foundation	Approvals, design, platform base	Approved SRS; architecture and threat model; database schema; cloud environment; authentication and role-based authorization; Super Admin dashboard foundation

Phase	Focus	Deliverables
2. Candidate and assessment management	Content and assignment	Candidate account and profile management; question bank; assessment creation, assignment, scheduling, configuration, and versioning
3. Candidate desktop application	Taking an assessment	Windows client; candidate login; assigned assessment screen; sequential questions; server-controlled timer; Submit and Next; auto-submission; session recovery; application logging
4. Evaluation and reporting	Results	Objective evaluation; negative marking; result generation; candidate reports; statistics; exports; audit-log review
5. Coding and SQL	Technical assessment	Code editor; language support; isolated execution service; hidden test cases; resource limits; SQL sandbox; coding scoring; status reporting
6. Lockdown and hardening	Security	Managed Windows kiosk configuration; candidate account restrictions; device registration; escape-path testing; penetration testing; dependency scanning; security logging; incident procedures
7. Production readiness	Go-live	Load testing; backup and restore testing; monitoring and alerts; deployment documentation; administrator guide; candidate instructions; user acceptance testing; production approval

18. Final Deliverables

No.	Deliverable
1	Software Requirements Specification
2	System Architecture Document
3	Database Design and ER Diagram
4	API Specification
5	Super Admin Web Dashboard
6	Candidate Desktop Application
7	Backend API and Assessment Services
8	Coding Execution Service
9	SQL Assessment Environment
10	Audit and Reporting Module
11	Windows Kiosk Deployment Configuration
12	Automated Unit and Integration Tests
13	Security and Performance Test Reports
14	Deployment and Recovery Documentation
15	Super Admin User Manual
16	Candidate User Instructions
17	Source Code and Build Instructions

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19. Conclusion

VeriSync is intended to be a secure, centralized, and extensible platform for technical placement recruitment. It combines administrator-controlled candidate management, configurable assessments, independent question timers, automatic submission, coding evaluation, reporting, and managed lab restrictions.

The central principle is that the candidate application is never trusted to enforce critical assessment rules. The backend controls identity, authorization, deadlines, answer acceptance, evaluation, and results, and operating-system kiosk restrictions complement application controls.

Production readiness is determined by verified acceptance criteria, security testing, performance testing, recovery testing, and formal stakeholder approval.

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20. Approval and Sign-Off

Approval confirms that the listed requirements form an agreed baseline for implementation. Subsequent changes shall be documented and reviewed.

Role	Name	Signature
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